

Ioannis Ziogas

Department of Biomedical Engineering

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in [Ioannis Ziogas](#)

🔗 gianniszgs.github.io

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OBJECTIVE

Build an independent academic career leading research that joins signal processing theory with AI for reliable representation learning, from biomedical signals, speech and healthcare to the internals of language models and beyond.

EDUCATION

Ph.D. in Biomedical Engineering

August 2022 – July 2026

Khalifa University of Science and Technology, Abu Dhabi, United Arab Emirates

- *Dissertation:* “Structure-aware Self- and Unsupervised Representation Learning for Acoustic and Physiological Time Series”
- *Advisors:* Prof. Ahsan H. Khandoker, Prof. Leontios J. Hadjileontiadis, Dr. Aamna Al Shehhi
- *GPA:* 3.84/4
- KU was ranked 147th globally in the QS World University Rankings 2027

M.Sc. (Integrated) in Electrical and Computer Engineering

September 2016 – April 2022

School of Engineering, Aristotle University of Thessaloniki (AUTH), Thessaloniki, Greece

Major: Telecommunications Engineering

- *Grade:* 7.83/10 (Upper Second-Class Honors – Very Good)
 - *Thesis:* “Classification and Characterization of the Effect of Migraine through Functional Connectivity Characteristics: Application to EEG Recordings from a Multimorbid Clinical Sample” (*April 2021 – April 2022*)
 - *Advisors:* Prof. Leontios J. Hadjileontiadis, Dr. Herbert F. Jelinek
 - AUTH is the second-highest ranking university in Greece
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RESEARCH EXPERIENCE

University of Toronto

August 2025 – July 2026

Graduate Visiting Researcher

The Edward S. Rogers Sr. Department of Electrical and Computer Engineering

Identity, Privacy, and Security Institute (IPSI)

Research Advisor: Prof. Dimitrios Hatzinakos

- Design of multi-modal self-supervised representation learning models to address omni-modality and full-stack representations in wearable time series processing.
- Application in activity recognition and biometric identification in wearable environments.
- Application in multi-modal wearable data synthesis.

Khalifa University of Science and Technology

August 2022 – July 2026

Doctoral Research

- Specialize in self-supervised and unsupervised learning algorithms for time-series data.
- Create novel structure-aware learning algorithms by integrating prior knowledge from advanced signal processing as inductive biases.
- Develop variational decomposition autoencoders, a theoretical and experimental framework to measure disentanglement in time series.
- Design of speech recognition, speech emotion recognition, dysarthria severity evaluation, physiological emotion recognition, activity recognition, identity recognition workflows.
- Hands-on experience on pre-training and fine-tuning variational autoencoder models, large acoustic models and state-of-the-art self-supervised learning models for audio and biomedical signals modalities.

- Design of personalization-oriented brain connectivity algorithms for EEG recordings.
- Detected migraine in a multimorbid clinical cohort and examined in-depth the electrophysiological hallmarks and phenotype of migraine.
- Innovatively integrate advanced signal processing techniques into functional brain connectivity measurement workflows.

- Performed in-depth analysis of rest tremor accelerometer recordings from Parkinson's Disease patients using advanced signal processing and machine learning techniques.
- Extensively validated different advanced signal processing techniques (higher order spectra, cepstral analysis) with advanced statistical significance techniques (surrogate data, statistical tests) and machine learning algorithms.
- Proposed new strategy in characterization of the tremor producing system in Parkinson's with respect to levodopa and deep brain stimulation interventions.

TEACHING EXPERIENCE

- **Aristotle University of Thessaloniki**, Signal Processing and Biomedical Technology Unit, Greece (June 2025 – Present)
MSc Thesis Supervision/Mentorship
 - *Candidates:* Ms. Maria N. Forou and Ms. Andriani Pappa, School of Electrical and Computer Engineering
 - *Supervisors:* Dr. Efstratia Ganiti Roumeliotou, **Dr. Ioannis Ziogas**, Prof. Leontios Hadjileontiadis
 - *Title:* "Development of a Smartphone-Based Serious Game Platform for supporting patients diagnosed with Psoriatic Arthritis"
 - Development of a comprehensive serious game platform for mobile phone interactions by grip and hand gestures using the smartphones' inbuilt electromagnetic sensor signal. Signal processing of the electromagnetic signal for grip strength and gesture recognition feature extraction. Game front-end and back-end design and game database design. Machine learning models for grip strength and gesture recognition prediction. Integration of the above components in a serious game platform where patients can exercise grip strength and fine motor skills.
 - Conceptualization and formation of thesis topic, student selection through interviews, research and technical guidance on signal processing and machine learning experiments.
- **Aristotle University of Thessaloniki**, Signal Processing and Biomedical Technology Unit, Greece (June 2025 – Present)
MSc Thesis Supervision/Mentorship
 - *Candidate:* Mr. Kyriakos Tsokkos, School of Electrical and Computer Engineering
 - *Supervisors:* Dr. Efstratia Ganiti Roumeliotou, **Dr. Ioannis Ziogas**, Prof. Leontios Hadjileontiadis
 - *Title:* "Affective Game Design with Emotionally Adaptive Avatars"
 - Explores affective game environments with multi-modal physiological and game log recordings from an emotionally adaptive perspective by using short-term signal analysis and graph neural networks modeling.
 - Conceptualization and formation of thesis topic, student selection through interviews, research and technical guidance on signal processing and machine learning experiments.
- **Aristotle University of Thessaloniki**, Signal Processing and Biomedical Technology Unit, Greece (June 2025 – June 2026)
MSc Thesis Supervision/Mentorship
 - *Candidate:* Ms. Maria Dimou, School of Electrical and Computer Engineering
 - *Supervisors:* **Dr. Ioannis Ziogas**, Dr. Efstratia Ganiti Roumeliotou, Prof. Leontios Hadjileontiadis
 - *Title:* "Learning Joint Multimodal Representations within Diverse Affective Game Environments for Development of a Sensor-less Physiological Monitoring Tool"
 - Explored unimodal and multi-modal autoencoder models for learning joint representations of physiological, inertial, and game log recordings. Cross-modality retrieval and generation was explored towards a signal-to-game and game-to-signal recording tool, in affective gaming environments with physiological monitoring.
 - Conceptualization and formation of thesis topic, student selection through interviews, research and technical guidance on signal processing and machine learning experiments.
- **Khalifa University of Science and Technology, Abu Dhabi, UAE** (Jan. 2026 – Jun. 2026)
Teaching Assistant: Fundamentals of Biosignal Processing (Biomedical Engineering 352, undergraduate)
 - Graded homework, quizzes and exams and held office hours

- **Khalifa University of Science and Technology, Abu Dhabi, UAE** (Jan. 2025 – Jun. 2025)
Teaching Assistant: Fundamentals of Biosignal Processing (Biomedical Engineering 352, undergraduate)
 - Graded homework, quizzes and exams and held office hours
- **Khalifa University of Science and Technology, Abu Dhabi, UAE** (Aug. 2024 – Dec. 2024)
Teaching Assistant: Biomedical Circuits and Signals (Biomedical Engineering 351, undergraduate)
 - Graded homework, quizzes and exams and held office hours
- **Khalifa University of Science and Technology, Abu Dhabi, UAE** (Jan. 2024 – Jun. 2024)
Teaching Assistant: Fundamentals of Biosignal Processing (Biomedical Engineering 352, undergraduate)
 - Graded homework, quizzes and exams and held office hours
- **Khalifa University of Science and Technology, Abu Dhabi, UAE** (Jan. 2024 – Jun. 2024)
Teaching Assistant: Biosensors: Fundamentals and Analysis (Biomedical Engineering 641, M.Sc. level)
 - Delivered lecture on “Digital Biomarkers in the AI and Big Data Era”
- **Khalifa University of Science and Technology, Abu Dhabi, UAE** (Aug. 2023 – Dec. 2023)
Teaching Assistant: Biomedical Circuits and Signals (Biomedical Engineering 351, undergraduate)
 - Graded homework, quizzes and exams and held office hours
- **Khalifa University of Science and Technology, Abu Dhabi, UAE** (Jan. 2023 – Jun. 2023)
Teaching Assistant: Biosensors: Fundamentals and Analysis (Biomedical Engineering 641, M.Sc. level)
 - Delivered lecture on “Digital Biomarkers: Analysis through Signal Processing and AI”
 - Delivered lab sessions on “Biosensors Data Analysis with Origin”

PUBLICATIONS

- **Ziogas, I.;** Khazaei E.; Taha, B.; AlShehhi A.; Khandoker, A. H.; Hadjileontiadis, L.J.; and Hatzinakos, D.; “Omni-modal decomposition autoencoders learn full-stack wearable disentangled representations” *Under revision at IEEE Journal of Biomedical and Health Informatics*, **2026**.
- **Ziogas, I.;** Khandoker, A.H.; AlShehhi, A.; Hadjileontiadis, L.J.; “Variational decomposition autoencoding improves disentanglement of latent representations”, *Under revision at npj Artif. Intell.*, **2026**.
- Alkhodari, M.; **Ziogas, I.;** Singh, J.; Grigolini, P.; Muschert, G.W.; Kusmartsev, F.; and Jelinek, H.F.; “Continuous bidirectional coupling of heart rate variability and emotions in dyad speaker-listener dynamics reveals unique interpersonal synchronization”. *Frontiers in Systems Neuroscience*, 20, p.1802891, **2026**.
- Alsaafeen, N. †; **Ziogas, I.;** †; Alsaedi, S.; Alshehhi, M.; Almheiri, S.; Rehman, A.; Al Shehhi, K.; Saleh, H.; Khandoker, A.; Pitsalidis, C.; Khawaja, A.; “Systematic Benchmarking of a Noise-Tolerant Conductive Hydrogel Electrode for Epidermal Bioelectronics”. *Advanced Science*, 13(20), p.e15131, **2026**. (†: co-first authorship)
- Alkhodari, M.; **Ziogas, I.;** Singh, J.; Grigolini, P.; Muschert, G.; Kusmartsev, F.; Jelinek, H. F.; “Bidirectional coupling of physiology and emotions reveals unique speaker–listener dynamics.”, *Social Cognitive and Affective Neuroscience*, 21(1), nsag019, **2026**.
- Bushara, F.; **Ziogas, I.;** Kumar, R.; Elkaffas, R.A.; Garg, A.; Samad, Y.A.; Hadjileontiadis, L.J.; Tardy, B.L.; Pitsalidis, C.; "Bioderived and all-solution-processed tribolayer component enables adaptive design of flexible nanocellulosic triboelectric nanogenerators." *Journal of Materials Chemistry A*, **2026**.
- Ganiti-Roumeliotou, E.; Apostolidis, G.; **Ziogas, I.;** Dias, S. B.; & Hadjileontiadis, L. J; “Impact of Dynamic Serious Game Environments on Multimodal Physiological Signals: A Multivariate Swarm-Prey Approach”. In *2025 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)* (pp. 1-7). IEEE, **2025**.
- Alhussein, G.; Alkhodari, M.; **Ziogas, I.;** Lamprou, C.; Khandoker, A. H.; Hadjileontiadis, L. J; . Exploring emotional climate recognition in peer conversations through bispectral features and affect dynamics. *Computer Methods and Programs in Biomedicine*, 265, 108695, **2025**.
- Alhussein, G.†; **Ziogas, I.;**†; Saleem, S.†; Hadjileontiadis, L.J.; “Speech Emotion Recognition in Conversations Using Artificial Intelligence: A Systematic Review and Meta-Analysis”, *Artificial Intelligence Review* 58, no. 7: 198, **2025**. (†: co-first authorship)
- **Ziogas, I.;** Hadjileontiadis, L.J.; Khandoker, A.H; AlShehhi, A.; “Self-supervised Graph Representation Learning for In-the-Wild Wearable and Smartphone based Emotion Recognition”, *accepted at IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, **2025**.
- Ganiti-Roumeliotou E.; **Ziogas, I.;** Dias, S.B.; Alhussein G.; Jelinek, H.F.; Hadjileontiadis, L.J.; “Beyond the Game: Multimodal Emotion Recognition Before, During, and After Gameplay”, *46th Annual International IEEE Engineering in Medicine and Biology Conference (EMBC)*, **2024**.

- **Ziogas, I.;** Alfalahi, H.; Khandoker, A.H.; Hadjileontiadis, L.J.; “CochCeps-Augment: A Novel Self-Supervised Contrastive Learning Using Cochlear Cepstrum-based Masking for Speech Emotion Recognition”, *IEEE International Conference on Acoustics, Speech, and Signal Processing Workshops (ICASSPW)*, **2024**.
- **Ziogas, I.†;** Lamprou, C.†; Hadjileontiadis, L.J.; “Bispectral Analysis of Parkinsonian Rest Tremor: New Characterization and Classification Insights Pre-/Post- DBS and Medication Treatment”, *IEEE Access PP(99):1-1*, **2023**. (†: co-first authorship)
- **Ziogas, I.†;** Leta, V.†; Lamprou, C.; Trivedi, D.; Zinzalias, P.; Staunton, J.; Odin, P.; Chaudhuri, K.R.; Charisis, V.; Hadjidimitriou, S.; Stouraitis, T.; Hadjileontiadis, L.J.; “Dynamic Monitoring of Probiotics Effect in Parkinson’s Disease Patients via Swarm Decomposition and Bispectral Analysis of Electrogastrograms”, *45th Annual International IEEE Engineering in Medicine and Biology Conference (EMBC)*, **2023**. (†: co-first authorship)
- Ganiti-Roumeliotou, E.†; **Ziogas, I.†;** Lamprou, C.†; Alhussein, G.; Alfalahi, H.; AlShehhi, A.; Dias, S.B.; Jelinek, H.F.; Stouraitis, T.; Hadjileontiadis, L.J.; “Classification of Children with ADHD through Task-related EEG recordings via Swarm-Decomposition-based Phase Locking Value”, *45th Annual International IEEE Engineering in Medicine and Biology Conference (EMBC)*, **2023**. (†: co-first authorship)
- Alhussein, G.; Alkhodari, M.; Lamprou, C.; **Ziogas, I.;** Ganiti-Roumeliotou, E.; Khandoker, A.H.; Hadjileontiadis, L.J.; “Emotional Climate Recognition in Conversations using Peers’ Speech-based Bispectral Features and Affect Dynamics”, *45th Annual International IEEE Engineering in Medicine and Biology Conference (EMBC)*, **2023**.
- Alfalahi, H.; AlShehhi, A., Lamprou, C.; **Ziogas, I.;** Ganiti-Roumeliotou, E.; Khandoker, A.H., Hadjileontiadis, L.J.; “Parkinsonian Tremor Detection with Compact Convolutional Transformer from Bispectrum Representation of tri-Axial Accelerometer Signals”, *45th Annual International IEEE Engineering in Medicine and Biology Conference (EMBC)*, **2023**.
- Lamprou, C.†; **Ziogas, I.†;** Ganiti-Roumeliotou, E.†; Alfalahi, H.; Alhussein, G.; AlShehhi, A., L.J.; “Deep Bispectral Image Analysis for IMU-based Parkinsonian Tremor Detection”, *IEEE 20th International Symposium on Biomedical Imaging (ISBI)*, **2023**. (†: co-first authorship)

JOURNAL AND CONFERENCE REVIEWING EXPERIENCE

- **Parkinsonism and Related Disorders** (Nov. 2025)
Reviewed 1 article
- **IEEE Transactions on Communications** (Nov. 2024)
Reviewed 1 article
- **IEEE-EMBS International Conference on Biomedical and Health Informatics** (Aug. 2024)
Reviewed 4 articles
- **IEEE Sensors Conference**(Jul. 2024)
Reviewed 6 articles
- **IEEE Biomedical Circuits and Systems Conference** (Jun. 2024)
Reviewed 1 article

PRESENTATIONS

- **Ziogas, I.;** Hadjileontiadis, L.J.; Khandoker, A.H.; AlShehhi, A.; “Self-supervised graph representation learning for in-the-wild wearable and smartphone based emotion recognition”, Poster Presentation, *IEEE ICASSP, Hyderabad International Convention Centre (HICC), Hyderabad, India, 2025*.
- **Ziogas, I.;** Alfalahi, H.; Khandoker, A.H.; Hadjileontiadis, L.J.; “CochCeps-Augment: A Novel Self-Supervised Contrastive Learning Using Cochlear Cepstrum-based Masking for Speech Emotion Recognition”, Poster Presentation, *IEEE ICASSP, COEX Exhibition Center, Seoul, Republic of Korea, 2024*.
- **Ziogas, I.;** Leta, V.; Lamprou, C.; Trivedi, D.; Zinzalias, P.; Staunton, J.; Odin, P.; Chaudhuri, K.R.; Charisis, V.; Hadjidimitriou, S.; Stouraitis, T.; Hadjileontiadis, L.J.; “Dynamic Monitoring of Probiotics Effect in Parkinson’s Disease Patients via Swarm Decomposition and Bispectral Analysis of Electrogastrograms”, Oral Presentation, *IEEE EMBC, International Convention Center, Sydney Harbour, Australia, 2023*.

- Alhussein, G.; Alkhodari, M.; Lamprou, C.; **Ziogas, I.**; Ganiti-Roumeliotou, E.; Khandoker, A.H.; Hadjileontiadis, L.J.; “Emotional Climate Recognition in Conversations using Peers’ Speech-based Bispectral Features and Affect Dynamics”, Poster Presentation, *IEEE EMBC, International Convention Center, Sydney Harbour, Australia, 2023.*

HONORS / AWARDS

Student Paper Competition Geographical Finalist

July 2023

*45th Annual International Conference of the IEEE Engineering in Medicine and Biology Society
Sydney, Australia*

Scientific Research Society Honor

June 2022 – June 2025

Associate Membership, Sigma Xi Scientific Research Society

Nominators: Dr. Ronald J. Swatzyna & Dr. Alexandra Roark, Rice University/Texas Medical Center Chapter

EXPERIMENTAL SKILLS

- **Self-supervised and unsupervised representation learning:**
 - Proficient in application of self-supervised pre-training schemes for image and time-series modalities, such as, contrastive learning and knowledge distillation (teacher-student) learning.
 - Proficient in the innovative conceptualization of self-supervised pre-training schemes for time-series modalities, by exploiting inductive biases from the advanced signal processing field.
 - Proficient in full stack representation analysis and appropriate metrics for unimodal and multi-modal representations: generative/reconstruction evaluation, downstream prediction tasks, disentanglement evaluation.
 - Proficient in the understanding of large acoustic models, and the incorporation of aforementioned self-supervised techniques in the pre-training of these models.
 - Proficient in analyzing representational properties of self-supervised and autoencoder-based learned embeddings.
 - Knowledgeable in transfer learning/fine-tuning of self-supervised learned models.
- **Advanced signal processing and analysis:**
 - Proficient in higher order statistics and spectra e.g. bispectrum, bicoherence, bicepstrum.
 - Proficient in applying signal decomposition techniques, and understanding their working principles, i.e., peak detection algorithms, frequency clustering, decomposition-specific components of variational mode decomposition, empirical mode decomposition, empirical wavelet transform, swarm decomposition, cochlear decomposition.
 - Proficient in classic signal processing analysis; spectral analysis, time-frequency and wavelet analysis, cepstral analysis.
 - Proficient in general and signal-specific pre-processing techniques (Speech, Electrocardiogram, Electroencephalogram, Accelerometer, Electrodermal Activity, Electromagnetic Sensor).
 - Proficient in classic and advanced functional connectivity / information transfer measures (non-directional, uni-directional, bi-directional).
- **Machine learning and deep learning analysis of time series, image and graph modalities:**
 - Proficient in deep learning analysis using state-of-the-art neural network architectures (convolutional, transformers, graph).
 - Proficient in deep network and classifier optimization techniques for hyperparameter tuning.
 - Proficient in classification, regression, clustering, dimensionality reduction, manifold learning, generative/reconstruction tasks.
 - Proficient in designing and evaluating machine learning loops, with cross-validation, feature selection, and feature explainability properties.
- **Speech recognition and analysis:**
 - Proficient in low-level speech analysis in the frequency domain using advanced signal processing techniques (time-frequency analysis, decomposition analysis, cepstral analysis).
 - Proficient in phoneme recognition, speaker identification.
 - Knowledgeable on inner workings of textless speech recognition systems e.g. wav2vec2, HuBERT, as well as language model driven speech recognition systems e.g. wav2vec2 + LM.

- Knowledgeable on disentangled representations of speech signals.
- **Data Analysis / Statistical Analysis:**
 - Proficient in univariate and multivariate statistical techniques; hypothesis testing, distribution fitting, dimensionality reduction techniques.
 - Knowledgeable on tabular data preprocessing techniques; imputation, feature selection.
 - Probability theory, generative probabilistic modeling.
- **Literature Systematization and Meta-Analysis:**
 - Proficient in conducting a systematic literature review, from the initial design/ research question setting phase, to the systematic search and screening of results, until the final knowledge distillation, according to widely accepted guidelines such as the PRISMA-DTA statement.
 - Proficient in meta-analytic techniques, such as, evaluating methodological quality and risk of bias, and designing random-effects models.
- **Biomedical Engineering:**
 - Knowledgeable on manifestations of emotion in physiological, speech and smartphone-based modalities.
 - Knowledgeable on Parkinson’s disease motor and non-motor manifestations and symptoms, and electrophysiological signatures. Knowledgeable on clinical scales used for measurement of symptoms, and translation of computational analysis results to clinically relevant insights.
 - Knowledgeable on electrophysiological and brain network dynamics of migraine and its comorbid neurological/mental disorders.

COMPUTATIONAL SKILLS

- Proficient in programming with Python
- Proficient in PyTorch
- Proficient in HuggingFace (audio/speech processing)
- Proficient in AI-assisted coding and workflow creation using LLMs (GitHub Copilot)
- Proficient in Weights & Biases – training and development record tracker
- Proficient in Sci-kit Learn, NumPy, Pandas
- Proficient in MATLAB
- Proficient with EEGLab MATLAB-plugin
- Experience with Tensorflow
- Experience with OriginLab
- Proficient in LaTeX
- Proficient in Microsoft Office: Word, Excel, PowerPoint
- Proficient with Mendeley reference manager, Rayyan screening tool

AFFILIATIONS & ACADEMIC ACTIVITIES

IEEE Signal Processing Society (SPS)
IEEE Engineering in Medicine and Biology Society (EMBS)

December 2020 - Present

Council Member, Secretary General

May 2020 – June 2021

IEEE-EMBS Student Chapter, AUTH

- Managed along with fellow council members to attract 40 new chapter members and raise interest about the biomedical engineering field
- Collaborated with university faculty and faculty from other institutions for the organization of biomedical engineering events, talks and workshops

Project Manager

May 2020 – June 2021

IEEE-EMBS Student Chapter, AUTH

- Managed student chapter’s undergraduate research project titled ”Serious Games for Anxiety” in a group of 20 chapter members
- Presented team and project outcomes during “Open student chapter day” in AUTH, December 2020.

- Organized literature review of state-of-the-art interventions for anxiety management and serious game design
- Organized and managed within-chapter workshops focused on biomedical signal processing (ECGs, GSRs) and machine learning techniques for analyzing biomedical data
- Collaborated on and discussed about project-related issues with university faculty

Conference Organizing Committee Member - DSAI 2024

November 2024

Khalifa University, Abu Dhabi

- 11th International Conference on Software Development and Technologies for Enhancing Accessibility and Fighting Info-exclusion
- Organized conference program, registration logistics and registration desk, and onsite logistics, for more than 100 participants